

Assessment Book — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12

Questions

Total marks 74

Q1. (4 marks) Two point charges lie on the x-axis: $q_1 = +3.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $q_2 = -5.0 \mu\text{C}$ at $x = +0.40 \text{ m}$. Using Coulomb's law with $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, calculate the magnitude of the electrostatic force on q_1 due to q_2 and give its direction (specify along $+x$ or $-x$, or 'towards q_2 /away from q_2 ').

Working:

Q2. (1 mark) A thin rod of length L lies along the x-axis from $x = -L/2$ to $x = +L/2$ and carries a uniform linear charge density λ (C/m). Find the magnitude of the electric field at point P on the y-axis at $y = a$ ($a > 0$). Show steps: start from Coulomb's law for a charge element $dq = \lambda dx$ and integrate. Give the result in terms of λ , L , a , and ϵ_0 and state the direction of the field.

- A. $E = (1 / (4\pi\epsilon_0)) * (\lambda L) / (a \sqrt{a^2 + (L/2)^2})$, directed along $+y$
- B. $E = (1 / (4\pi\epsilon_0)) * (2\lambda a) / (a^2 + (L/2)^2)$, directed along $+y$
- C. $E = (1 / (4\pi\epsilon_0)) * (\lambda L) / (a^2 + (L/2)^2)$, directed along $+y$
- D. $E = (1 / (4\pi\epsilon_0)) * (2\lambda) / a * [1 - a / \sqrt{a^2 + (L/2)^2}]$, directed along $+y$

Q3. (3 marks) For each of the three charge distributions below, state a suitable Gaussian surface, justify why it matches the symmetry, and give the magnitude and direction of the electric field (in terms of the given charge parameters and ϵ_0). Do this briefly for: (a) an infinite plane sheet with uniform surface charge density σ ; (b) an infinitely long straight wire with uniform linear charge density λ ; (c) a thin uniformly charged spherical shell of radius R and total charge q for the regions $r > R$ and $r < R$. (Total 3 marks: 1 mark each.)

Q4. (6 marks) A permanent electric dipole consists of two point charges $+q$ and $-q$ separated by distance $2a$. The dipole is placed in a uniform electric field E that makes an angle θ with the line joining the charges (the dipole axis).

(a) Define the dipole moment vector p for this dipole, including its magnitude and direction. (1 mark)

(b) Derive an expression for the torque τ acting on the dipole in this uniform field. Give the magnitude and the vector form, and state the direction of the torque using the right-hand rule. (3 marks)

(c) Explain whether the dipole experiences a net force in this uniform field, and how your answer would change if the field were non-uniform. In your explanation, briefly address the common misconception that a neutral object attracted to a charged object must have gained net charge. (2 marks)

Q5. (4 marks) A small insulating sphere carries a net charge of $-4.80 \times 10^{-19} \text{ C}$. (a) Calculate the number of excess electrons on the sphere. (b) The sphere is then touched to an identical neutral conducting sphere and the two spheres are separated. Calculate the final charge on each sphere and the (approximate) number of excess electrons on each. Use the elementary charge magnitude $e = 1.602 \times 10^{-19} \text{ C}$. Show units and steps.

Working:

Q6. (1 mark) Two point charges lie on the x-axis: $+4.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $-2.0 \mu\text{C}$ at $x = 0.30 \text{ m}$. What is the net electric field at the point $x = 0.20 \text{ m}$? Give magnitude and direction. (Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$; carry units through.)

- A. $2.70 \times 10^6 \text{ N/C}$, directed in the $+x$ direction
- B. $8.99 \times 10^5 \text{ N/C}$, directed in the $+x$ direction
- C. $1.80 \times 10^6 \text{ N/C}$, directed in the $+x$ direction
- D. $2.70 \times 10^6 \text{ N/C}$, directed in the $-x$ direction

Q7. (3 marks) A thin insulating spherical shell of radius R carries a total charge $+Q$ uniformly distributed over its surface. In words (no calculus), describe the electric field E (magnitude, direction, and dependence on distance r) for each region: (a) outside the shell ($r > R$), (b) on the surface ($r = R$), and (c) inside the shell ($r < R$). Briefly state the reason for each behaviour.

Q8. (6 marks) Explain how electric charges behave differently in conductors and in insulators. In your answer: (a) give a clear definition of a conductor and of an insulator in terms of electron mobility; (b) describe where excess charge resides when a solid conducting sphere is given extra electrons and when a solid insulating rod is given extra electrons; and (c) give one everyday example of a conductor and one everyday example of an insulator, naming the material and a common use.

Q9. (4 marks) Two point charges lie on the x-axis in vacuum. Charge $Q_1 = +3.0 \mu\text{C}$ is at $x = 0.00 \text{ m}$ and charge $Q_2 = -1.0 \mu\text{C}$ is at $x = 0.20 \text{ m}$. Calculate the electric field vector (magnitude and direction) at the point P located at $x = 0.10 \text{ m}$. Give your answer in N C^{-1} and state the direction along the x-axis.

Working:

Q10. (1 mark) A fixed charge $+Q$ sits at the origin and creates an electric field. A small positive test charge $+q$ is brought close to $+Q$ to measure the field. Which statement correctly describes the roles of $+Q$ and $+q$ when defining the electric field of $+Q$?

- A. $+Q$ is the source charge that produces the electric field; $+q$ is the test charge that experiences the force. $+q$ must be small (or fixed so it does not disturb $+Q$) so the field

being measured is that of +Q alone.

- B. Both +Q and +q are equally source charges that produce separate electric fields; the measured field at a point is just the field of +q at that point because it is closer.
- C. A test charge must be exactly zero to avoid changing the field; therefore +q should be taken to zero for the electric field of +Q to exist.
- D. The electric field is a property of the test charge +q and is measured by observing how +Q moves in response to +q.

Q11. (3 marks) Two point charges, +q and -q, are separated by a small distance. On the diagram below, draw the electric field lines for this charge pair and then, in words, state three properties of electric field lines that explain (a) the direction of the arrows you drew, (b) what the spacing (density) of the lines represents, and (c) whether the absolute number of lines drawn is a unique physical property of the field.

(Answer must include the sketch and three short statements.)

Q12. (4 marks) Two point charges +q and -q are placed on the x-axis at $x = +a$ and $x = -a$ respectively. Find an expression for the electric field magnitude E on the +x-axis at a point $x = r$ where $r \gg a$. Using a binomial expansion to first order in a/r , show that the field falls off as $1/r^3$ and express your result in terms of the dipole moment $p = q(2a)$. State the direction of the field. (4 marks)

Working:

Q13. (1 mark) A small flat surface has area magnitude dS and unit normal $\hat{n}$. A uniform electric field E passes nearby. Which expression correctly gives the electric flux through this small area element?

- A. $\Phi = E dS \cos\theta$, where θ is the angle between E and the surface plane
- B. $\Phi = (E \cdot \hat{n}) dS$, where $\hat{n}$ is the unit normal to the area
- C. $\Phi = E \times dS$, the cross product of the electric field and the area element
- D. $\Phi = E dS \sin\theta$, where θ is the angle between E and the unit normal $\hat{n}$

Q14. (3 marks) Two point charges lie on the x-axis: $q_1 = +3.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $q_2 = -2.0 \mu\text{C}$ at $x = +0.50 \text{ m}$. Using Coulomb's law, calculate (a) the magnitude of the electrostatic force on q_1 due to q_2 and (b) the direction of that force. Give the force as a magnitude with units and state its direction along the x-axis.

Q15. (4 marks) A thin straight rod of length L lies on the x-axis from $x = 0$ to $x = L$ and carries a uniform positive linear charge density λ (C m^{-1}). Find the electric field vector E at point P on the x-axis located at $x = -a$ ($a > 0$). Give the magnitude and direction of E in terms of λ , L , a and the constant $k = 1/(4\pi\epsilon_0)$.

Working:

Q16. (1 mark) For each charged object below, name the simplest Gaussian surface to use and

give the magnitude of the electric field at the specified point. Show the relation between flux and enclosed charge that leads to the formula. (i) Infinite plane with uniform surface charge density σ : field at a point a short distance from the plane. (ii) Infinitely long straight wire with linear charge density λ : field at radial distance r from the wire. (iii) Thin spherical shell of radius R carrying total charge q : field at (a) a point outside the shell at distance $r > R$ and (b) a point inside at distance $r < R$.

- A. Plane: use a small cylindrical pillbox whose flat faces are parallel to the plane; $E = \sigma/(2\epsilon_0)$ directed normal to the plane. Wire: use a coaxial cylindrical surface of radius r and length L ; $E = \lambda/(2\pi \epsilon_0 r)$ radially outward. Shell: use a spherical surface centered on the shell; for $r > R$, $E = q/(4\pi \epsilon_0 r^2)$ radial, and for $r < R$, $E = 0$.
- B. Plane: use a spherical Gaussian surface crossing the plane; $E = \sigma/\epsilon_0$ because both sides contribute. Wire: use a rectangular box around a short segment; $E = \lambda/(\pi \epsilon_0 r)$ pointing outward. Shell: use a cylindrical surface coaxial with the shell; for $r > R$, $E = q/(2\pi \epsilon_0 r^2)$ and inside $E = q/(4\pi \epsilon_0 R^2)$.
- C. Plane: use a pillbox with faces parallel to the plane; $E = \sigma/(2\epsilon_0)$ normal to the plane. Wire: use a coaxial cylindrical Gaussian surface of radius r and length L ; $E = \lambda/(4\pi \epsilon_0 r)$ radial. Shell: use a spherical Gaussian surface centred on the shell; for $r > R$, $E = q/(4\pi \epsilon_0 r^2)$ and for $r < R$, $E = 0$.
- D. Plane: use a pillbox with faces parallel to the plane; $E = \sigma/(2\epsilon_0)$ normal to the plane. Wire: use a coaxial cylindrical Gaussian surface of radius r and length L ; $E = \lambda/(2\pi \epsilon_0 r)$ radial. Shell: use a spherical Gaussian surface centred on the shell; for $r > R$, $E = q/(4\pi \epsilon_0 r^2)$ and for $r < R$, $E = q/(4\pi \epsilon_0 R^2)$.

Q17. (3 marks) A permanent electric dipole consists of charges $+q$ and $-q$ separated by distance $2a$. The dipole moment vector p points from $-q$ to $+q$. The dipole is placed in a uniform electric field E . (a) Write the vector expression for the torque τ on the dipole and give its magnitude in terms of p , E and the angle θ between p and E . (b) State whether the dipole feels a net force in (i) a uniform electric field and (ii) a non-uniform electric field, and briefly explain why in each case.

Q18. (4 marks) Two small conducting spheres A and B are isolated from the surroundings. Sphere A carries a net charge of $+8.0 \times 10^{-19} \text{ C}$ and sphere B carries a net charge of $-4.8 \times 10^{-19} \text{ C}$. The two spheres are touched together and then separated. Using the charge quantisation relation $q = n e$ with $e = 1.60 \times 10^{-19} \text{ C}$, calculate (a) the final net charge on each sphere (in coulombs) and (b) the number n of excess electrons on each sphere after separation. State the sign of n and explain briefly what a positive or negative n means physically. (4 marks)

Working:

Q19. (1 mark) Two identical point charges $+3.0 \mu\text{C}$ are fixed on the x -axis at $x = 0.00 \text{ m}$ and $x = 0.20 \text{ m}$. What is the net electric field (magnitude and direction) at the midpoint $x = 0.10 \text{ m}$?

- A. Zero newtons per coulomb (0 N/C) at the midpoint
- B. $2.7 \times 10^6 \text{ N/C}$ pointing to the right (positive x)
- C. $2.7 \times 10^6 \text{ N/C}$ pointing to the left (negative x)
- D. $5.4 \times 10^6 \text{ N/C}$ pointing to the right (positive x)

Q20. (4 marks) Two point charges lie on the x -axis in vacuum: charge $q_1 = +3.0 \mu\text{C}$ at $x = 0.00 \text{ m}$

and charge $q_2 = -1.0 \mu\text{C}$ at $x = 0.20 \text{ m}$. Calculate the electric field vector (magnitude and direction) at the point $x = 0.10 \text{ m}$. Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. (4 marks)

Working:

Q21. (1 mark) A small metal sphere and a small plastic sphere each receive an added $+2.0 \mu\text{C}$ (microcoulombs) of charge at a single point on their surfaces. After the charges settle, where is the added charge located on each object?

- A. On the metal sphere the added $+2.0 \mu\text{C}$ stays stuck near the point where it was placed; on the plastic sphere the charge spreads uniformly over the outer surface.
- B. On the metal sphere the added $+2.0 \mu\text{C}$ spreads over the outer surface; on the plastic sphere the added $+2.0 \mu\text{C}$ remains localized near the point where it was placed.
- C. On both the metal and the plastic spheres the added $+2.0 \mu\text{C}$ spreads uniformly over their outer surfaces.
- D. On both the metal and plastic spheres the added $+2.0 \mu\text{C}$ remains trapped at the point of application and does not move.

Q22. (4 marks) A single point charge $Q = +5.0 \mu\text{C}$ is fixed at the origin. Calculate the electric field vector at the point P located at $x = +0.20 \text{ m}$ on the x-axis. Give the magnitude (in N C^{-1}) and the direction. (Take $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.)

Working:

Q23. (4 marks) A point source charge $Q = +5.0 \mu\text{C}$ is fixed at the origin. (a) Calculate the electric field magnitude produced by Q at a distance $r = 0.20 \text{ m}$ from the origin. (b) A test charge $q = +1.0 \text{ nC}$ is placed at that location. Calculate the electric force on q and state its direction. (c) Briefly explain why q is called a test charge and why its magnitude should be small compared with Q when defining the electric field.

(Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Show all steps and include units.)

Working:

Q24. (4 marks) Two isolated positive point charges, A and B, are drawn with field-line diagrams. Around charge A eight equally spaced field lines leave the charge; around charge B four equally spaced field lines leave the charge. Point P is at distance r from charge A along one of its field lines. Point Q is at the same distance r from charge B along one of its field lines. (a) Calculate the ratio of the magnitudes of the electric fields $|E_P| : |E_Q|$ at P and Q. (b) Give a brief physical explanation (one or two sentences) that justifies this ratio and addresses whether the absolute number of drawn field lines is a fixed physical property of the charge.

(Mark allocation: 2 marks for the numeric ratio, 2 marks for the explanation.)

Working:

Answer Key — teacher copy, not for distribution to students

Answer Key

Q1.

Answer: Magnitude = 8.4×10^{-1} N (0.84 N). Direction = +x (to the right), i.e. towards q2 (attractive).

Working: Step 1 — Write Coulomb's law for magnitude: $F = k |q_1 q_2| / r^2$.

Step 2 — Substitute numbers (convert μC to C): $q_1 = 3.0 \times 10^{-6}$ C, $q_2 = 5.0 \times 10^{-6}$ C (use absolute value for magnitude). $r = 0.40$ m.

Calculate product of charges: $|q_1 q_2| = (3.0 \times 10^{-6} \text{ C})(5.0 \times 10^{-6} \text{ C}) = 15.0 \times 10^{-12} \text{ C}^2 = 1.50 \times 10^{-11} \text{ C}^2$.

Compute numerator $k|q_1 q_2|$: $(8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(1.50 \times 10^{-11} \text{ C}^2) = 8.99 \times 1.50 \times 10^{(9-11)} \text{ N}\cdot\text{m}^2 = 13.485 \times 10^{-2} \text{ N}\cdot\text{m}^2 = 0.13485 \text{ N}\cdot\text{m}^2$.

Compute r^2 : $(0.40 \text{ m})^2 = 0.1600 \text{ m}^2$.

Divide to get force magnitude: $F = 0.13485 \text{ N}\cdot\text{m}^2 / 0.1600 \text{ m}^2 = 0.8428 \text{ N} \sim 8.4 \times 10^{-1} \text{ N}$ (0.84 N).

Step 3 — Determine direction: q_1 is positive and q_2 is negative, so the force is attractive; q_1 is pulled toward q_2 which lies at +x. Therefore the force on q_1 is along +x (to the right), i.e. towards q_2 .

Units are newtons (N).

Rubric — Correct numerical magnitude (with units) and correct direction of the force on q_1 .

Full credit (4 marks): Gives the calculated magnitude ~ 0.84 N with correct units AND states the direction as +x (or 'towards q_2 ' / 'to the right'). Calculation shows Coulomb's law with charges converted to coulombs and r in metres, and arithmetic consistent with working.

Partial credit (2 marks): Gives a numerically correct magnitude (within about $\pm 5\%$, e.g. 0.80–0.89 N) but either omits units or gives an unclear direction (e.g. just says 'attractive' without specifying toward +x or toward q_2), OR gives correct direction but numeric magnitude wrong by a clear arithmetic error (factor-of-two type) but correct setup shown.

No credit (0 marks): Shows no correct numerical magnitude (wrong formula or no valid substitution) and gives incorrect or missing direction.

Q2.

Correct: A. $E = (1 / (4\pi\epsilon_0)) * (\lambda L) / (a \sqrt{a^2 + (L/2)^2})$, directed along +y

A. Correct. Start with $dq = \lambda dx$. The y-component from dq is $dE_y = (1/(4\pi\epsilon_0)) (dq) a / (x^2 + a^2)^{3/2}$. Integrate x from $-L/2$ to $+L/2$; symmetry gives twice the integral from 0 to $L/2$. Evaluating integral $dx/(x^2 + a^2)^{3/2} = x/(a^2 \sqrt{x^2 + a^2})$ yields $E = (1/(4\pi\epsilon_0)) \lambda L / (a \sqrt{a^2 + (L/2)^2})$. Direction is +y because all contributions point upward.

B. This expression has wrong algebra from the integral. A student choosing it likely tried to combine powers without performing the correct antiderivative; the numerator $2\lambda a$ looks like from $2 \cdot (\lambda a)$ but the denominator should be a times a square root, not a quadratic sum without the correct power dependence. This choice reflects mis-evaluating the integral $\int dx/(x^2 + a^2)^{3/2}$.

C. This option misses the square root in the denominator that comes from the antiderivative. A student selecting it probably replaced $\sqrt{a^2 + (L/2)^2}$ by $(a^2 + (L/2)^2)$ incorrectly, i.e. lost a power of $1/2$ when simplifying the

integral, yielding an extra factor of length in the denominator and wrong units.

D. This formula resembles the field from a finite charged segment written in terms of endpoint angles, but the bracketed combination is incorrect for this geometry. Picking it indicates the student tried to use an angular result but made an algebraic error when converting endpoints to x-coordinates; it does not reduce to the correct integral result and has incorrect dependence on a and L .

Q3.

Answer: (a) Infinite plane sheet, surface charge density σ :

Gaussian surface: a short cylindrical pillbox with flat faces parallel to the plane and equal area A , symmetric about the plane. Symmetry/justification: the plane is infinite and uniform, so E is perpendicular to the plane and has the same magnitude on both sides; flux only through the two flat faces. Field: $E = \sigma/(2\epsilon_0)$ directed normal to the plane on each side.

(b) Infinitely long straight wire, linear charge density λ :

Gaussian surface: a coaxial cylindrical surface of radius r and length L . Symmetry/justification: translational and cylindrical symmetry imply E is radial, same magnitude at all points at distance r , and perpendicular to curved surface; no flux through end caps. Field: $E(r) = \lambda/(2\pi \epsilon_0 r)$ directed radially outward (or inward if $\lambda < 0$).

(c) Thin uniformly charged spherical shell radius R , total charge q :

Gaussian surface: concentric spherical surfaces of radius r . Symmetry/justification: spherical symmetry makes E radial and same at all points on a sphere. For $r > R$ (outside): enclosed charge = q , so $E(r) = q/(4\pi \epsilon_0 r^2)$ directed radially outward. For $r < R$ (inside): enclosed charge = 0, so $E = 0$.

Rubric — Choice and justification of Gaussian surface; correct expression and direction of E for each part.

Full marks (3) (3 marks): Each part (a), (b), (c) names an appropriate Gaussian surface, gives a one-sentence symmetry justification (why E is uniform and its direction), and states the correct magnitude and direction of E : (a) $E = \sigma/(2\epsilon_0)$ normal to plane; (b) $E = \lambda/(2\pi \epsilon_0 r)$ radial; (c) outside $E = q/(4\pi \epsilon_0 r^2)$ radial, inside $E = 0$.

Partial credit (2) (2 marks): Two of the three parts correctly give an appropriate Gaussian surface with brief symmetry justification and correct field expression and direction; the third part has either a correct surface but incorrect/omitted justification or an incorrect/blank field expression.

Minimal/incorrect (0) (0 marks): Fewer than two parts correct: wrong Gaussian surfaces or no symmetry justification and incorrect or missing field expressions for two or more parts.

Q4.

Answer: (a) The electric dipole moment vector $\mathbf{p}$ has magnitude $p = q \times 2a$ and points from the negative charge ($-q$) to the positive charge ($+q$). (1 mark)

(b) Each charge experiences a force $F = qE$ (direction depends on sign). The forces on $+q$ and $-q$ are equal in magnitude and opposite in direction if the field is uniform, but because they act at different positions they produce a torque. The torque on the dipole about its centre can be written as the cross product

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}.$$

Magnitude: $|\boldsymbol{\tau}| = p E \sin\theta = (2aq) E \sin\theta$.

Vector/direction: τ is perpendicular to the plane containing p and E ; using the right-hand rule for $p \times E$, point fingers along p (from $-q$ to $+q$), curl toward E ; the thumb gives the direction of τ . (3 marks)

(c) In a uniform electric field the net force on the dipole is zero because the forces on $+q$ and $-q$ are equal in magnitude and opposite in direction and their vector sum cancels. However, in a non-uniform (spatially varying) electric field the magnitudes of the forces on the two charges will differ, producing a net force on the dipole. For a small dipole in a slowly varying field this net force is approximately $(p \cdot \text{grad})E$ (i.e. depends on the gradient of the field).

Addressing the misconception: attraction of a neutral object to a charged object (for example paper to a comb) does not mean the neutral object gained net charge. The neutral object can become polarized (an induced dipole): charges shift slightly inside it so that near-side induced opposite charge experiences a stronger attraction than the farther-side repulsion, producing net attraction without any net charge transfer. (2 marks)

Rubric — Accuracy and completeness of dipole moment definition, torque derivation (magnitude and vector), direction statement, and force behavior in uniform vs non-uniform fields (including misconception clarification).

Full credit (6 marks): All parts correct: p defined as $p = 2aq$, direction from $-q$ to $+q$ (1 mark); torque given in vector form $\tau = p \times E$ and magnitude $|\tau| = pE \sin\theta$ with substitution $p = 2aq$ (3 marks); clear statement that net force is zero in a uniform field with correct reasoning and explanation that a non-uniform field produces a net force (e.g. depends on field gradient), plus correct clarification that induced polarization (not net charging) explains attraction of neutral objects (2 marks).

Partial credit (4 marks): Mostly correct but incomplete or minor omission: correct p and torque magnitude (or vector) but missing or unclear statement of torque direction or missing right-hand rule (lose 1 mark); or states net force is zero in uniform field but gives only brief or slightly incorrect reasoning or omits non-uniform-field explanation (score 3–4 marks total).

Minimal credit (1 marks): Only fragments correct: e.g. quotes $\tau = pE \sin\theta$ without defining p properly, or gives incorrect sign/direction for torque, or merely states that non-uniform fields can give a force without correct reasoning; misconception not addressed or explained incorrectly. (award 0–2 marks depending on amount of correct content present).

Q5.

Answer: (a) Number of excess electrons on the charged sphere: $n = q / (-e) = (-4.80 \times 10^{-19} \text{ C}) / (-1.602 \times 10^{-19} \text{ C}) = 2.995 \sim 3$ electrons. So the sphere has 3 excess electrons. (b) Total charge is conserved and equals $-4.80 \times 10^{-19} \text{ C}$. After touching two identical spheres, charge is shared equally, so charge on each $= (-4.80 \times 10^{-19} \text{ C}) / 2 = -2.40 \times 10^{-19} \text{ C}$. Number of excess electrons on each (approximate) $= (-2.40 \times 10^{-19} \text{ C}) / (-1.602 \times 10^{-19} \text{ C}) = 1.498 \sim 1.5$ electrons. Because charge is quantised in whole electrons, in reality one sphere would have 2 excess electrons and the other 1 excess electron (total 3 electrons), but the equal sharing calculation gives $-2.40 \times 10^{-19} \text{ C}$ per sphere (about 1.5 electrons each) as the ideal continuous result.

Working: (a) Given $q = -4.80 \times 10^{-19} \text{ C}$ and $e = 1.602 \times 10^{-19} \text{ C}$ (magnitude of electron charge).

Compute number of excess electrons n :

$$n = q / (-e) = (-4.80 \times 10^{-19} \text{ C}) / (-1.602 \times 10^{-19} \text{ C})$$

First divide magnitudes: $4.80 / 1.602 = 2.995$ (unitless).

So $n = 2.995 \sim 3$ electrons. Units: electrons (count).

(b) Conservation of charge: total $Q_{\text{total}} = -4.80 \times 10^{-19} \text{ C}$.

Two identical spheres share charge equally, so charge on each $Q_{\text{final}} = Q_{\text{total}} / 2 = (-4.80 \times 10^{-19} \text{ C}) / 2 = -2.40 \times 10^{-19} \text{ C}$ (per sphere).

Number of excess electrons on each (approximate, using continuous division): $n_{\text{each}} = Q_{\text{final}} / (-e) = (-2.40 \times 10^{-19} \text{ C}) / (-1.602 \times 10^{-19} \text{ C}) = 2.40 / 1.602 = 1.498 \sim 1.5$ electrons.

Note on quantisation: electrons are discrete, so the actual integer distribution consistent with total 3 electrons is one sphere with 2 electrons and the other with 1 electron, giving charges $-3.204 \times 10^{-19} \text{ C}$ and $-1.602 \times 10^{-19} \text{ C}$ respectively; the equal-sharing ($-2.40 \times 10^{-19} \text{ C}$ each) is the ideal continuous result.

Rubric — Correct computation and interpretation of number of electrons; correct conservation and equal sharing of charge; acknowledgement of quantisation if non-integer electron result.

Full marks (4 marks): All steps shown with units. Correctly computes $n = 2.995 \sim 3$ electrons for part (a); correctly computes final charge on each sphere as $-2.40 \times 10^{-19} \text{ C}$ and obtains ~ 1.50 electrons each for part (b). Explicitly states that because electrons are discrete, the actual integer distribution would be one sphere with 2 electrons and the other with 1 electron (total 3).

Partial credit (2 marks): Correct method shown with units but with an arithmetic slip (e.g., gives $n \sim 2$ or 4, or gives final charge magnitude correct but wrong electron division), or computes equal sharing correctly but omits the note on quantisation. Work shows understanding of $q = n e$ and conservation of charge.

No credit (0 marks): Incorrect or missing calculations; does not use $q = n e$ or conservation of charge; no units shown or entirely wrong answers.

Q6.

Correct: A. $2.70 \times 10^6 \text{ N/C}$, directed in the +x direction

A. Correct: Calculate each field at $x = 0.20 \text{ m}$. From $+4.0 \mu\text{C}$ at 0.00 m : $r_1 = 0.20 \text{ m}$, $E_1 = k \cdot 4.0 \times 10^{-6} / (0.20^2) = 8.99 \times 10^5 \text{ N/C}$ to +x (away from positive charge). From $-2.0 \mu\text{C}$ at 0.30 m : $r_2 = 0.10 \text{ m}$, $E_2 = k \cdot 2.0 \times 10^{-6} / (0.10^2) = 8.99 \times 10^6 \text{ N/C}$ toward the negative charge, which at the point is also +x. Sum: $E_{\text{net}} = 8.99 \times 10^5 + 1.798 \times 10^6 = 2.697 \times 10^6 \sim 2.70 \times 10^6 \text{ N/C}$ to +x.

B. This option gives only the field from the $+4.0 \mu\text{C}$ charge ($E_1 = 8.99 \times 10^5 \text{ N/C}$ to +x) and ignores the larger contribution from the $-2.0 \mu\text{C}$ charge at 0.30 m ($E_2 \sim 1.798 \times 10^6 \text{ N/C}$). A student choosing this is adding only one source instead of using superposition.

C. This option equals approximately the field from only the $-2.0 \mu\text{C}$ charge ($E_2 \sim 1.798 \times 10^6 \text{ N/C}$). A student selecting it has used only the closer negative charge's field and omitted the contribution from the $+4.0 \mu\text{C}$ charge, violating superposition.

D. This option has the correct magnitude but the wrong direction. A student choosing it likely misapplied the direction rule: both fields at $x = 0.20 \text{ m}$ point to the +x side (away from the positive charge and toward the negative charge), so the net cannot be to -x.

Q7.

Answer: (a) Outside ($r > R$): The field is radially outward and has magnitude $E = (1/(4\pi\epsilon_0)) \cdot Q / r^2$ (units N C^{-1}). The shell behaves like a point charge at its centre because of spherical symmetry (Gauss's law).

(b) On the surface ($r = R$): The field is radially outward with magnitude $E = (1/(4\pi\epsilon_0)) \cdot Q / R^2$ (N C^{-1}); this is the value from the outside expression evaluated at $r = R$.

(c) Inside ($r < R$): The field is zero ($E = 0 \text{ N C}^{-1}$) everywhere inside the hollow shell. By spherical symmetry and Gauss's law, any Gaussian surface inside encloses no net charge, so the net flux

and therefore the field must be zero.

Rubric — Correct description of the electric field in the three regions (outside, on surface, inside) with appropriate dependence on r and brief justification.

Full marks (3 marks): All three regions correctly described: (a) outside — $E = (1/4\pi\epsilon_0) \cdot Q/r^2$, radially outward, with statement that shell acts like point charge by symmetry/Gauss's law; (b) on surface — $E = (1/4\pi\epsilon_0) \cdot Q/R^2$, radially outward; (c) inside — $E = 0$, with brief Gauss's-law/symmetry justification.

Partial understanding (2 marks): Two of the three regions correct OR expressions for all three regions given correctly but justification missing or only partially stated (e.g. gives E values but no mention of symmetry or Gauss's law).

Insufficient (0 marks): Fewer than two regions correct, or major errors in formulas/directions, and no valid justification provided.

Q8.

Answer: A conductor is a material in which valence electrons (or charge carriers) can move freely through the material; an insulator is a material in which electrons are tightly bound to atoms and cannot move freely. When a solid conducting sphere receives extra electrons, the electrons move freely and redistribute themselves over the outer surface of the sphere until electrostatic equilibrium is reached; there is no excess charge left in the interior and the electric field inside the conductor is zero. By contrast, when a solid insulating rod receives extra electrons (for example by rubbing), the extra electrons stay localized near the region where they were placed and do not move freely to the whole surface; the charge remains stuck in that part of the insulator. An everyday example of a conductor is copper (used for electrical wiring because it allows current to flow easily). An everyday example of an insulator is plastic (used to coat wires and make handles because it prevents charge flow).

Rubric — Accuracy and completeness of definitions, description of charge distribution in conductor and insulator, and provision of correct everyday examples.

Full credit (6 marks): Provides correct definitions for conductor and insulator (electron mobility stated); correctly describes that excess charge on a conducting sphere moves to the outer surface (interior field zero) and that excess charge on an insulating rod remains localized where placed; gives one correct everyday example of a conductor with use and one correct everyday example of an insulator with use.

Partial credit (3 marks): Gives partially correct definitions (mentions mobility but incomplete); correctly describes either the conductor charge behavior (surface redistribution) or the insulator behavior (localized charge) but not both, or gives correct behaviors but omits the interior field comment; provides at least one correct example (material and plausible use) or gives both examples but one is incorrect.

Minimal or no credit (0 marks): Gives incorrect or very vague definitions (no mention of electron mobility), fails to describe where excess charge resides for conductor and insulator, and/or gives no correct everyday examples.

Q9.

Answer: The net electric field at P is $2.7 \times 10^5 \text{ N C}^{-1}$ directed to the right (positive x -direction).

Working: Constants and conversions:

Permittivity $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$; $k = 1/(4\pi \epsilon_0) = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.

Convert charges: $Q_1 = +3.0 \mu\text{C} = 3.0 \times 10^{-6} \text{ C}$; $Q_2 = -1.0 \mu\text{C} = -1.0 \times 10^{-6} \text{ C}$.

Positions: $x_1 = 0.00 \text{ m}$, $x_P = 0.10 \text{ m}$, $x_2 = 0.20 \text{ m}$.

Distance from Q_1 to P: $r_1 = x_P - x_1 = 0.10 \text{ m}$ (field due to Q_1 at P points away from $+Q_1$, i.e. to the right, $+x$).

Magnitude $E_1 = k |Q_1| / r_1^2 = (8.99 \times 10^9)(3.0 \times 10^{-6}) / (0.10)^2$

$$E_1 = (8.99 \times 10^9 \times 3.0 \times 10^{-6}) / 0.01$$

$$E_1 = (26.97 \times 10^3) / 0.01 = 2.697 \times 10^6 \text{ N C}^{-1} \rightarrow 2.70 \times 10^6 \text{ N C}^{-1} \text{ to three significant figures, direction } +x.$$

Distance from Q2 to P: $r_2 = x_2 - x_P = 0.20 - 0.10 = 0.10 \text{ m}$ (field due to Q2 at P: Q2 is negative so field points toward Q2, i.e. to the right as well, $+x$).

$$\text{Magnitude } E_2 = k |Q_2| / r_2^2 = (8.99 \times 10^9)(1.0 \times 10^{-6}) / (0.10)^2$$

$$E_2 = (8.99 \times 10^3) / 0.01 = 8.99 \times 10^5 \text{ N C}^{-1} \rightarrow 8.99 \times 10^5 \text{ N C}^{-1}.$$

Both fields point in the $+x$ direction, so add magnitudes:

$$E_{\text{net}} = E_1 + E_2 = 2.697 \times 10^6 + 8.99 \times 10^5 = (2.697 + 0.899) \times 10^6 = 3.596 \times 10^6 \text{ N C}^{-1}.$$

Round to two significant figures appropriate to input: $3.60 \times 10^6 \text{ N C}^{-1}$ to the right.

(If using three significant figures consistent with the charges: $3.60 \times 10^6 \text{ N C}^{-1}$ to the $+x$ direction.)

Note: Depending on rounding conventions, an answer of about $3.6 \times 10^6 \text{ N C}^{-1}$, $+x$, earns full credit.

Rubric — Correct calculation of electric field magnitudes from both charges, correct vector directions, and correct net field (magnitude and direction) with units.

Full credit (4 marks): All steps shown: correct use of $E = kQ/r^2$ for each charge with distances $r_1 = 0.10 \text{ m}$ and $r_2 = 0.10 \text{ m}$, correct directions (both fields $+x$), correct addition to give net $E_{\text{net}} \sim 3.6 \times 10^6 \text{ N C}^{-1}$ directed $+x$. Units stated.

Partial credit (2 marks): Correct calculation for both magnitudes but error in direction (e.g. one field given as $-x$) or arithmetic error that yields a net within about a factor of 2 of correct value; or only one field correctly calculated and the other attempted. Units included.

No credit (0 marks): Formula not used correctly or no meaningful calculation (e.g. missing distances, wrong dependence on r , or no units).

Q10.

Correct: A. $+Q$ is the source charge that produces the electric field; $+q$ is the test charge that experiences the force. $+q$ must be small (or fixed so it does not disturb $+Q$) so the field being measured is that of $+Q$ alone.

A. Correct. The source charge $+Q$ creates the field; the test charge $+q$ probes it by feeling a force $F = qE$. To define the field of $+Q$, $+q$ must be small or fixed so it does not significantly alter $+Q$ or the field configuration.

B. Wrong: this implies the test charge and source are interchangeable and that the field is dominated by $+q$ because of proximity. The test charge should be negligible so it does not produce a comparable field; the measured field is that due to $+Q$, not $+q$.

C. Wrong: a test charge need not be zero; it must be small enough that its own field is negligible. Taking $q \rightarrow 0$ is a limiting concept for defining $E = F/q$, but physically we use a small nonzero q that does not disturb the source.

D. Wrong: the electric field is a property of the source charge(s) and space, not of the test charge. The test charge experiences the field (force $F = qE$); it does not 'have' the field being measured.

Q11.

Answer: Sketch: Field lines start at the positive charge ($+q$) and end at the negative charge ($-q$); lines are more curved near the space between the charges and symmetric in a dipole shape. Arrows point away from $+q$ and toward $-q$.

Three properties (one per letter):

(a) Direction: The tangent to a field line at any point gives the direction of the electric field there, so arrows point away from the positive charge and toward the negative charge.

(b) Spacing (density): The local density of field lines represents the relative magnitude of the electric field; where lines are closer together the field is stronger, and where they are farther

apart the field is weaker.

(c) Number of lines: The absolute number of lines drawn is not a unique physical property — one may draw more or fewer lines to illustrate strength; only the relative density and direction have physical meaning.

Rubric — Correct sketch and three accurate statements about (a) direction, (b) density representing field strength, and (c) non-uniqueness of absolute line count.

3 marks (3 marks): Sketch shows lines beginning at +q and ending at -q with arrows pointing away from +q toward -q (dipole-like shape). Statement (a) correctly says tangent gives field direction and arrows point away from +q/toward -q. Statement (b) correctly says local line density indicates relative field strength (closer lines = stronger field). Statement (c) correctly says the absolute number of drawn lines is not an intrinsic physical property and may be chosen for illustration.

2 marks (2 marks): Sketch is mostly correct (lines generally from +q to -q) and two of the three statements are correct. For example, (a) and (b) correct but (c) missing or incorrect, or (a) correct while (b) and (c) provide partly correct ideas (e.g. mention density but not its meaning).

0 marks (0 marks): Sketch is incorrect (e.g. arrows pointing toward +q or lines crossing) or fewer than two of the three required statements are correct. Statements that assert the absolute number of lines is a fixed physical quantity or confuse direction/density are scored zero.

Q12.

Answer: For $r \gg a$ the electric field on the +x-axis is directed along +x and has magnitude $E \sim (1/4\pi\epsilon_0) (2p) / r^3$

where $p = q(2a)$. Thus E falls off as $1/r^3$ (units N C^{-1}).

Working: 1. Exact fields from each charge at point $x = r$ ($r > a$):

E_+ from +q at $x = +a$ points in +x direction with magnitude $E_+ = (1/4\pi\epsilon_0) q / (r - a)^2$.

E_- from -q at $x = -a$ points in +x direction (since -q attracts a positive test charge toward -x, but located at -a gives field at r directed toward -a, i.e. -x; however because the charge is negative the field at r is toward -a which is in the -x direction; careful: using signs explicitly below gives correct net direction).

2. Use algebra with signed fields (taking +x direction positive):

Field from +q: $E_+ = (1/4\pi\epsilon_0) q / (r - a)^2$ (positive).

Field from -q: distance to -q is $r + a$, and field points toward -q, i.e. in the -x direction, so $E_- = -(1/4\pi\epsilon_0) q / (r + a)^2$.

Net field: $E = E_+ + E_- = (1/4\pi\epsilon_0) q [1/(r - a)^2 - 1/(r + a)^2]$.

3. Combine the difference of squares:

$$1/(r - a)^2 - 1/(r + a)^2 = [(r + a)^2 - (r - a)^2] / [(r - a)^2 (r + a)^2]$$

$$\text{Numerator: } (r^2 + 2ar + a^2) - (r^2 - 2ar + a^2) = 4ar.$$

$$\text{So the difference} = 4ar / [(r - a)^2 (r + a)^2].$$

$$\text{Thus } E = (1/4\pi\epsilon_0) q (4ar) / [(r - a)^2 (r + a)^2].$$

4. For $r \gg a$, approximate $(r \pm a)^2 \sim r^2(1 \pm a/r)^2 \sim r^2(1 \pm 2a/r)$ to first order, but product $(r - a)^2(r + a)^2 \sim (r^2 - a^2)^2 \sim r^4(1 - a^2/r^2)^2 \sim r^4$ to leading order. Therefore to leading order

$$E \sim (1/4\pi\epsilon_0) q (4ar) / r^4 = (1/4\pi\epsilon_0) (4qa) / r^3.$$

5. Write in terms of the dipole moment $p = q(2a)$: then $4qa = 2p$, so

$$E \sim (1/4\pi\epsilon_0) (2p) / r^3.$$

6. Direction: the sign of E found above is positive, so the field is along +x (away from the dipole on the positive axis). Units are N C^{-1} , showing the field scales as $1/r^3$.

(End of working.)

Rubric — Correct derivation showing $1/r^3$ dependence and expressing the result in terms of dipole moment p

Full credit (4 marks): Correct algebraic setup for fields from $+q$ and $-q$, correct combination to get $E = (1/4\pi\epsilon_0) q[1/(r-a)^2 - 1/(r+a)^2]$, correct expansion or approximation for $r \gg a$ leading to $E \sim (1/4\pi\epsilon_0)(4qa)/r^3$, substitution $p = q(2a)$ to give $E \sim (1/4\pi\epsilon_0)(2p)/r^3$, and statement that E is along $+x$; units indicated. (4 marks)

Partial credit (2 marks): Correct setup and algebra to the combined difference but either a) approximates denominator incompletely giving the right $1/r^3$ scaling but incorrect numerical factor (e.g. missing factor 2), or b) gets the factor right but fails to express result clearly in terms of p ; direction may be missing. Shows understanding that field $\sim 1/r^3$. (2 marks)

No credit (0 marks): Incorrect setup of fields, wrong combination of signs/distances, or no valid approximation for $r \gg a$; result does not show $1/r^3$ dependence or is absent. (0 marks)

Q13.

Correct: B. $\Phi = (\mathbf{E} \cdot \hat{\mathbf{n}}) dS$, where $\hat{\mathbf{n}}$ is the unit normal to the area

A. Incorrect: If θ is the angle between $\mathbf{E}$ and the surface plane, then $\cos\theta$ equals the component of $\mathbf{E}$ parallel to the plane, not the component through the normal. This confuses the angle reference; the correct through-surface component uses the angle to the normal.

B. Correct: By definition the vector area element is $dS = \hat{\mathbf{n}} dS$, so the flux through the element is $\mathbf{E} \cdot dS = (\mathbf{E} \cdot \hat{\mathbf{n}}) dS$. Units: $E \text{ (N/C)} \times \text{area (m}^2\text{)}$ gives $\text{N}\cdot\text{m}^2/\text{C}$, and the dot product takes the component of $\mathbf{E}$ normal to the surface.

C. Incorrect: The cross product gives a vector perpendicular to both $\mathbf{E}$ and the area vector and does not measure the scalar flux through the surface. This mistakes perpendicular-area magnitude for the scalar dot-product flux.

D. Incorrect: $\Phi = E dS \sin\theta$ uses the sine of the angle to the normal, which gives the component of $\mathbf{E}$ perpendicular to the normal (i.e., parallel to the surface). This confuses sine and cosine for the normal component.

Q14.

Answer: Magnitude: 0.216 N. Direction: $+x$ (toward q_2). Explanation: Using Coulomb's law $F = k |q_1 q_2| / r^2$ with $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $q_1 = 3.0 \times 10^{-6} \text{ C}$, $q_2 = -2.0 \times 10^{-6} \text{ C}$ and $r = 0.50 \text{ m}$ gives $F = (8.99 \times 10^9)(3.0 \times 10^{-6})(2.0 \times 10^{-6})/(0.50^2) = 0.216 \text{ N}$. The charges are opposite signs so the force is attractive; since q_2 is at $+0.50 \text{ m}$, the force on q_1 is toward $+x$.

Working: Step 1 — write Coulomb's law (magnitude): $F = k |q_1 q_2| / r^2$, where $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

Step 2 — substitute values with units: $q_1 = 3.0 \times 10^{-6} \text{ C}$, $q_2 = -2.0 \times 10^{-6} \text{ C}$ (use absolute value for magnitude), $r = 0.50 \text{ m}$.

$F = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \times (3.0 \times 10^{-6} \text{ C}) \times (2.0 \times 10^{-6} \text{ C}) / (0.50 \text{ m})^2$.

Step 3 — compute numerator: $(8.99 \times 10^9) \times (6.0 \times 10^{-12} \text{ C}^2) = 8.99 \times 6.0 \times 10^{(9-12)} \text{ N}\cdot\text{m}^2 = 53.94 \times 10^{-3} \text{ N}\cdot\text{m}^2 = 0.05394 \text{ N}\cdot\text{m}^2$.

Step 4 — compute denominator: $(0.50 \text{ m})^2 = 0.25 \text{ m}^2$.

Step 5 — divide: $F = 0.05394 \text{ N}\cdot\text{m}^2 / 0.25 \text{ m}^2 = 0.21576 \text{ N} \sim 0.216 \text{ N}$.

Step 6 — direction: q_1 is positive and q_2 is negative $\rightarrow$ attraction. q_2 is at $+0.50 \text{ m}$ relative to q_1 at 0.00 m , so the force on q_1 is toward $+x$.

Final answer: $F = 0.216 \text{ N}$, directed $+x$ (toward q_2).

Rubric — Correct numerical magnitude with units and correct x -axis direction with justification.

Full credit (3 marks): Correct magnitude ($\sim 0.216 \text{ N}$, shown with units) calculated using $F = k |q_1 q_2| / r^2$ with substituted values and correct arithmetic, and correct direction given ($+x$ toward q_2) with explanation that opposite charges attract.

Partial credit (2 marks): Magnitude calculated and correct to within about 10% (answer between 0.195 N and 0.238 N) with working shown, but direction omitted or stated without linking to attraction/repulsion; OR correct

direction (+x) stated but magnitude incorrect by more than 10% though working shows an attempt to use Coulomb's law.

Minimal credit (1 marks): Shows the Coulomb's law formula and substitutions but contains algebraic or arithmetic errors leading to a magnitude far from the correct value, and direction either incorrect or not justified; or gives only the qualitative direction (attraction) with no numeric calculation.

Q15.

Answer: Electric field at P is directed toward the rod (i.e. in the +x direction if λ were negative; for positive λ the field points in the +x direction toward increasing x). The x-component is $E_x = k \lambda (1/a - 1/(a + L))$.

So the electric field vector is

$E = k \lambda (1/a - 1/(a + L)) \hat{i}$ (pointing in the +x direction for positive λ).

Working: Take an element of length dx at position x ($0 \leq x \leq L$) carrying charge $dq = \lambda dx$ (units C). The position of P is $x_P = -a$. The separation from dq to P is $r = x - (-a) = x + a$ (units m). The contribution to the electric field at P from dq , using Coulomb's law, is $dE = k dq / r^2$, directed from dq toward P (for positive λ the field points from the positive charge toward P, i.e. in the +x direction because dq is to the right of P). The field contributions are all along the x-axis, so vector signs reduce to a sign for the x-component. Using +x as the direction from P toward the rod, $dE_x = k (\lambda dx)/(x + a)^2$.

Integrate over the rod from $x = 0$ to $x = L$:

$$E_x = \int_0^L k \lambda (dx)/(x + a)^2.$$

Compute the integral: let $u = x + a$, $du = dx$, limits $u = a$ to $u = a + L$.

$$E_x = k \lambda \int_{u=a}^{u=a+L} u^{-2} du = k \lambda [-u^{-1}]_a^{a+L} = k \lambda (1/a - 1/(a + L)).$$

Units check: k ($N m^2 C^{-2}$) $\times \lambda$ ($C m^{-1}$) $\times (1/m)$ gives $N C^{-1} = V m^{-1}$, correct for electric field.

Therefore $E = E_x \hat{i} = k \lambda (1/a - 1/(a + L)) \hat{i}$, pointing in the +x direction for positive λ .

Rubric — Correct setup and evaluation of the integral for the x-component of E ; correct final expression and direction.

Full marks (4 marks): Correctly writes $dq = \lambda dx$, expresses the separation as $r = x + a$, sets up $E_x = \int k \lambda dx/(x + a)^2$ with limits 0 to L, evaluates the integral to obtain $E_x = k \lambda (1/a - 1/(a + L))$, and states the correct direction (+x for positive λ).

Partial credit (2 marks): Sets up the integral with $dq = \lambda dx$ and correct limits but makes an algebraic or sign error in integration or inverting limits, yielding an expression algebraically different but clearly from the same integral (e.g. incorrect sign or swapped denominators). Shows correct reasoning about direction or units.

Minimal credit (1 marks): Attempts to use Coulomb's law but fails to set up the integral properly (e.g. uses dq without λdx , uses incorrect separation r , or integrates over wrong limits) or gives only a dimensional or verbal statement without the correct formula; direction incorrect or not stated.

Q16.

Correct: A. Plane: use a small cylindrical pillbox whose flat faces are parallel to the plane; $E = \sigma/(2\epsilon_0)$ directed normal to the plane. Wire: use a coaxial cylindrical surface of radius r and length L ; $E = \lambda/(2\pi \epsilon_0 r)$ radially outward. Shell: use a spherical surface centered on the shell; for $r > R$, $E = q/(4\pi \epsilon_0 r^2)$ radial, and for $r < R$, $E = 0$.

A. Correct. For the infinite plane a pillbox gives flux through two faces: $2EA = Q_{enc}/\epsilon_0 = \sigma A/\epsilon_0 \Rightarrow E = \sigma/(2\epsilon_0)$. For the infinite wire a coaxial cylinder gives flux $\Phi = E(2\pi rL) = Q_{enc}/\epsilon_0 = \lambda L/\epsilon_0 \Rightarrow E = \lambda/(2\pi \epsilon_0 r)$. For the thin spherical shell a concentric sphere gives $\Phi = E(4\pi r^2) = q/\epsilon_0$ for $r > R \Rightarrow E = q/(4\pi \epsilon_0 r^2)$; if $r < R$ no charge is enclosed so $E = 0$.

B. Wrong. The student chose a spherical Gaussian for the plane and got $E = \sigma/\epsilon_0$; this double-counts because the

correct pillbox has two faces so each face sees $\sigma/(2\epsilon_0)$. The wire answer has the wrong coefficient ($\lambda/(\pi \epsilon_0 r)$) from an incorrect flux area (using π instead of 2π). The sphere answers use wrong surfaces and wrong factors so enclosed-charge application is incorrect.

C. Partially right for the plane and sphere outside, but wrong for the wire: $E = \lambda/(4\pi \epsilon_0 r)$ is half the correct value. That error comes from using half the cylindrical surface area (e.g., using $2\pi rL$ where $4\pi rL$ was mistakenly assumed) or halving the flux; the correct cylinder gives $2\pi rL$ so $E = \lambda/(2\pi \epsilon_0 r)$.

D. Plane and wire are mixed: the plane and wire formulas are partly correct in form but the shell-inside field is wrong. Saying E inside the shell equals $q/(4\pi \epsilon_0 R^2)$ implies a nonzero field independent of r , which violates Gauss's law because a concentric sphere inside encloses zero charge so E must be zero.

Q17.

Answer: (a) The torque on the dipole is given by the vector cross product $\tau = p \times E$. Its magnitude is $|\tau| = pE \sin\theta$, where θ is the angle between p and E . The torque tends to rotate the dipole so that p aligns with E . (b) (i) In a uniform electric field the net force on the dipole is zero because the forces on $+q$ and $-q$ are equal in magnitude and opposite in direction, producing only a torque. (ii) In a non-uniform electric field the dipole can experience a net force because the field strength at the positions of $+q$ and $-q$ is different; the unequal forces do not cancel and produce a net translational force (which tends to pull the dipole toward regions of stronger field if aligned appropriately).

Rubric — Correct expression and magnitude for torque; correct statements and brief explanations about net force in uniform and non-uniform fields.

Full credit (3 marks): States $\tau = p \times E$ and $|\tau| = pE \sin\theta$ with θ defined; explains clearly that in a uniform field net force = 0 because forces on $+q$ and $-q$ cancel and that in a non-uniform field a net force can occur because the field differs at the two charges (unequal forces do not cancel).

Partial credit (2 marks) (2 marks): Gives the correct torque formula (vector or magnitude) with θ , and correctly states that net force = 0 in a uniform field, but gives only a weak or incomplete explanation for the non-uniform case (e.g. says 'net force possible' without explaining unequal forces due to field gradient).

Partial credit (1 mark) (1 marks): Gives only the magnitude relation $|\tau|$ proportional to pE (or mentions torque tends to align p with E) or correctly states only one of the force conclusions (either uniform field = 0 or non-uniform field $\neq 0$) with minimal or no explanation.

Q18.

Answer: Total initial charge = $(+8.0 \times 10^{-19}) + (-4.8 \times 10^{-19}) = +3.2 \times 10^{-19}$ C. When touched, charge is conserved and shared equally, so final charge on each sphere = $(3.2 \times 10^{-19} \text{ C}) / 2 = +1.6 \times 10^{-19}$ C. Using $q = n e$ with $e = 1.60 \times 10^{-19}$ C gives $n = q / e = (+1.6 \times 10^{-19} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) = +1$. Therefore each sphere has net charge $+1.6 \times 10^{-19}$ C, corresponding to $n = +1$ (one elementary positive charge). A positive n means a deficit of one electron (one more proton's worth of positive charge than electrons); a negative n would mean an excess of electrons.

Working: Step 1: Compute total charge (conserve charge).

$$Q_{\text{total}} = Q_A + Q_B = +8.0 \times 10^{-19} \text{ C} + (-4.8 \times 10^{-19} \text{ C}) = +3.2 \times 10^{-19} \text{ C}.$$

Step 2: Charge shared equally when conductors touch; each gets half.

$$Q_{\text{each}} = Q_{\text{total}} / 2 = (3.2 \times 10^{-19} \text{ C}) / 2 = 1.6 \times 10^{-19} \text{ C}.$$

Step 3: Convert to number of elementary charges using $q = n e$, $e = 1.60 \times 10^{-19}$ C.

$$n = q / e = (1.6 \times 10^{-19} \text{ C}) / (1.60 \times 10^{-19} \text{ C}) = 1.0 \rightarrow n = +1 \text{ (dimensionless)}.$$

Units carried: charges in coulombs throughout; e in coulombs so n is unitless.

Interpretation: $n = +1$ means each sphere is missing one electron compared with being neutral (a net positive elementary charge).

Rubric — Correct final charge per sphere (C) and correct integer number of excess electrons n with clear use of $q = n e$ and sign interpretation.

Full credit (4 marks): Correct total and final charge: $Q_{\text{each}} = +1.6 \times 10^{-19} \text{ C}$, and correct calculation $n = +1$ with use of $q = n e$ shown and a correct physical interpretation of positive n (deficit of one electron). Units carried correctly.

Partial credit (2 marks): Final charge in coulombs correct ($Q_{\text{each}} = +1.6 \times 10^{-19} \text{ C}$) but the conversion to n is missing, non-integer, or incorrectly signed OR the student gives $n = 1$ without showing use of $q = n e$. Units largely present.

No credit (0 marks): Final charge incorrect and no correct use of $q = n e$, or arithmetic/misconception that charge is destroyed on contact (i.e. not conserved).

Q19.

Correct: A. Zero newtons per coulomb (0 N/C) at the midpoint

A. Correct—each charge produces an electric field of magnitude $E = kq/r^2$ with $r = 0.10 \text{ m}$: $E = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})/(0.10 \text{ m})^2 = 2.697 \times 10^6 \text{ N/C}$. At the midpoint the two fields are equal in magnitude and opposite in direction, so their vector sum is zero.

B. This choice treats the field from one charge as the net field. The student forgot to add vector directions: both charges produce equal-magnitude fields at the midpoint but they point in opposite directions and cancel, so the net is not a single $2.7 \times 10^6 \text{ N/C}$ to the right.

C. This choice has the correct single-charge magnitude but the wrong direction for the net field. It mistakes the field from one charge for the net; because the two equal fields are opposite, they do not leave a net $2.7 \times 10^6 \text{ N/C}$ to the left.

D. This choice doubles the single-charge magnitude as if the two fields simply add in the same direction. The misconception is treating magnitudes as scalars to sum without accounting for direction; here the two equal fields are opposite and cancel, so you should not get $5.4 \times 10^6 \text{ N/C}$.

Q20.

Answer: The net electric field at $x = 0.10 \text{ m}$ is $3.60 \times 10^6 \text{ N/C}$ directed in the $+x$ (right) direction.

Working: Step 1 — Distances and directions:

- Point of interest is at $x = 0.10 \text{ m}$.
- Distance to q_1 : $r_1 = 0.10 \text{ m}$ (from $x = 0.00$ to 0.10). Field from q_1 (positive) points away from q_1 , i.e. in $+x$ direction.
- Distance to q_2 : $r_2 = 0.10 \text{ m}$ (from $x = 0.20$ to 0.10). Field from q_2 (negative) points toward q_2 , i.e. also in $+x$ direction.

Step 2 — Magnitudes using $E = k|q|/r^2$:

- $E_1 = k * |q_1| / r_1^2 = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) * (3.0 \times 10^{-6} \text{ C}) / (0.10 \text{ m})^2$
 $= 8.99 \times 10^9 * 3.0 \times 10^{-6} / 0.01 \text{ N/C}$
 $= 8.99 \times 10^9 * 3.0 \times 10^{-4} \text{ N/C}$
 $= 2.697 \times 10^6 \text{ N/C}$.
- $E_2 = k * |q_2| / r_2^2 = (8.99 \times 10^9) * (1.0 \times 10^{-6}) / (0.10)^2$
 $= 8.99 \times 10^9 * 1.0 \times 10^{-4} \text{ N/C}$
 $= 8.99 \times 10^5 \text{ N/C}$.

Step 3 — Vector sum (both in +x direction):

$$- E_{\text{net}} = E_1 + E_2 = 2.697 \times 10^6 \text{ N/C} + 8.99 \times 10^5 \text{ N/C} = 3.596 \times 10^6 \text{ N/C}.$$

Step 4 — Report to three significant figures and state direction:

$$- E_{\text{net}} \sim 3.60 \times 10^6 \text{ N/C in the +x (right) direction.}$$

Rubric — Correct use of superposition to compute magnitudes of fields from each charge, correct vector sum and stated units/direction.

Full marks (4 marks): Correctly computes E_1 and E_2 with k and r^2 shown and units, adds them as vectors (noting both point +x), and gives final answer $3.60 \times 10^6 \text{ N/C}$ in the +x direction (or equivalent to three significant figures).

Partial credit (2 marks): Shows correct formulas and distances and gets either both individual magnitudes correct but makes an arithmetic rounding error in the final sum (award 2 marks), or obtains one individual field correctly and the other incorrect but shows correct direction for both fields (award 2 marks).

No credit (0 marks): Fails to use superposition, gives wrong directions for one or both fields, or gives an answer with no working, no units, or completely incorrect magnitude.

Q21.

Correct: B. On the metal sphere the added $+2.0 \mu\text{C}$ spreads over the outer surface; on the plastic sphere the added $+2.0 \mu\text{C}$ remains localized near the point where it was placed.

A. This reverses the behaviours of conductors and insulators. A conductor does not keep extra charge localized because free electrons move; the plastic (insulator) does not let charge spread freely.

B. Correct — metals are conductors: free electrons move until excess charge resides on the outer surface (measurable as $+2.0 \mu\text{C}$ distributed over the surface, in coulombs that is $+2.0 \times 10^{-6} \text{ C}$). Plastics are insulators: charges cannot move freely, so the added $+2.0 \mu\text{C}$ stays localized near the application point.

C. This choice assumes both materials allow charges to move freely. That is wrong because the plastic is an insulator and prevents free motion of charge so it does not generally form a uniform surface distribution.

D. This choice assumes neither material allows charge to move. That is incorrect because metals are conductors: free charges will move and redistribute to the surface (so the metal will not keep the $+2.0 \mu\text{C}$ stuck at the application point).

Q22.

Answer: Magnitude: $E = 1.12 \times 10^6 \text{ N C}^{-1}$. Direction: along the +x direction (away from the positive charge).

Working: 1. Write the electric field from a point charge: $E = kQ / r^2$, directed radially away from a positive charge.

2. Substitute values: $Q = +5.0 \mu\text{C} = 5.0 \times 10^{-6} \text{ C}$, $r = 0.20 \text{ m}$, $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$.

$$E = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}) \times (5.0 \times 10^{-6} \text{ C}) / (0.20 \text{ m})^2.$$

3. Calculate denominator: $(0.20 \text{ m})^2 = 0.0400 \text{ m}^2$.

$$\text{Numerator: } (8.99 \times 10^9) \times (5.0 \times 10^{-6}) = 44.95 \times 10^3 = 4.495 \times 10^4 \text{ (units: N}\cdot\text{m}^2\cdot\text{C}^{-1}).$$

4. $E = (4.495 \times 10^4) / 0.0400 \text{ N C}^{-1} = 1.12375 \times 10^6 \text{ N C}^{-1}$.

5. Round to three significant figures (given data to three s.f.): $E = 1.12 \times 10^6 \text{ N C}^{-1}$.

6. Direction: Q is positive, so the electric field at P ($x = +0.20 \text{ m}$) points away from the charge, i.e. in the +x direction.

Final answer: $E = 1.12 \times 10^6 \text{ N C}^{-1}$, directed along +x.

Rubric — Correct magnitude of the electric field with units, and correct direction.

Full credit (4 marks): Uses $E = kQ/r^2$ with $Q = 5.0 \times 10^{-6} \text{ C}$ and $r = 0.20 \text{ m}$, shows substitution and arithmetic,

obtains $E \sim 1.12 \times 10^6 \text{ N C}^{-1}$ with units, and states direction as +x (away from the positive charge).

Partial credit (2 marks): Shows the correct formula and substitution but has an arithmetic error or gives the magnitude without units or with one incorrect significant-figure step (result close to correct, e.g. within 20%), and correctly identifies the direction as away from the positive charge.

Minimal or no credit (0 marks): Uses an incorrect formula (e.g. missing r^2 or using F instead of E), or gives an incorrect direction (toward the charge), or no relevant working shown.

Q23.

Answer: (a) Electric field magnitude: $E = kQ/r^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(5.0 \times 10^{-6} \text{ C})/(0.20 \text{ m})^2 = 1.12 \times 10^6 \text{ N/C}$.

(b) Electric force on q : $F = qE = (1.0 \times 10^{-9} \text{ C})(1.12 \times 10^6 \text{ N/C}) = 1.12 \times 10^{-3} \text{ N}$, directed radially outward from the source charge (repulsive since both charges are positive).

(c) The charge q is called a test charge because it probes the electric field produced by the source charge Q without significantly altering that field. q should be small compared with Q so that its own field does not appreciably move or change the source charge Q or the field configuration; this ensures the field measured depends only on the source charge.

Working: (a) Compute E from a point charge formula, showing units stepwise:

$$Q = 5.0 \text{ } \mu\text{C} = 5.0 \times 10^{-6} \text{ C}$$

$$r = 0.20 \text{ m}$$

$$k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

$$E = kQ/r^2$$

$$= (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \times (5.0 \times 10^{-6} \text{ C}) / (0.20 \text{ m})^2$$

$$= (8.99 \times 10^9 \times 5.0 \times 10^{-6}) / 0.04 \text{ N/C}$$

$$= (8.99 \times 5.0 \times 10^3) / 0.04 \text{ N/C} \quad [\text{since } 10^9 \times 10^{-6} = 10^3]$$

$$= (44.95 \times 10^3) / 0.04 \text{ N/C}$$

$$= 1.12375 \times 10^6 \text{ N/C} \sim 1.12 \times 10^6 \text{ N/C}.$$

$$(b) q = 1.0 \text{ nC} = 1.0 \times 10^{-9} \text{ C}$$

$$F = qE = (1.0 \times 10^{-9} \text{ C})(1.12375 \times 10^6 \text{ N/C})$$

$$= 1.12375 \times 10^{-3} \text{ N} \sim 1.12 \times 10^{-3} \text{ N}.$$

Direction: Both Q and q are positive, so the Coulomb force is repulsive; the force on q is radially outward from the origin.

(c) Explanation: The electric field E is defined as the force per unit test charge ($E = F/q$) produced by the source charge Q . To measure or define this field without changing it, the test charge q must be small so that its own electric field and the force it exerts on the source charge are negligible; otherwise the presence of q would alter the field that we aim to measure.

Rubric — Correct calculation of field and force; correct direction; clear explanation of why q is a test charge and must be small.

Full marks (4 marks): All parts correct: (a) E computed correctly with units and correct numeric value $\sim 1.12 \times 10^6 \text{ N/C}$; (b) F computed correctly with units and correct numeric value $\sim 1.12 \times 10^{-3} \text{ N}$ and direction stated as radially outward (repulsive); (c) concise correct explanation that q probes the field, defined by $E = F/q$, and q must be small so it does not appreciably alter the source charge or the field.

Partial credit (2 marks): One computational part incorrect or missing but the other computation correct, and a

partially correct explanation given (mentions that q probes the field or that it should be small but without linking to not altering the source/field). Example: correct E but wrong arithmetic in F (or vice versa), or correct numbers but missing units or slightly wrong by a factor of 2.5 or less.

No credit (0 marks): Major errors in computations (orders of magnitude wrong), wrong direction, and no or incorrect explanation about the role and required smallness of the test charge.

Q24.

Answer: a) $|E_P| : |E_Q| = 2 : 1$ (i.e. $|E_P| = 2|E_Q|$).

b) The density of field lines leaving a charge represents the relative field strength; since A has twice as many lines leaving as B and the points are the same distance r , the field at P is twice that at Q. The absolute number of drawn field lines is not a fixed physical property — the artist can choose how many lines to draw, so only the relative density (lines per unit area around the charge) indicates relative field strength.

Working: Step 1: Use the rule that, in field-line diagrams, the local field magnitude is proportional to the local density of field lines (lines per unit area through a surface around the charge).

Step 2: At the same radial distance r , and along a single field line, the local density around A is proportional to the number of lines leaving A (8) and around B proportional to the number leaving B (4), assuming equal spacing.

Step 3: Ratio $|E_P|/|E_Q| = (\text{number of lines from A})/(\text{number of lines from B}) = 8/4 = 2$.

Therefore $|E_P| : |E_Q| = 2 : 1$, so $|E_P| = 2|E_Q|$.

Units: electric field in newtons per coulomb (N C^{-1}), but ratio is dimensionless.

Rubric — Correct numeric ratio and correct explanation linking ratio to field-line density and stating that absolute line count is not a fixed physical property

Full credit (4 marks): Gives the correct numeric ratio 2:1 ($|E_P| = 2|E_Q|$) AND provides a clear explanation that field magnitude is proportional to field-line density so A's twice-as-many lines at the same distance give twice the field; explicitly states that the absolute number of drawn lines is arbitrary (not an intrinsic physical property).

Partial credit (2 marks): Gives the correct numeric ratio 2:1 but explanation is incomplete or partly incorrect (e.g. says more lines mean stronger field but does not mention density or that absolute number is arbitrary), OR gives correct qualitative explanation but numeric ratio is wrong.

No credit (0 marks): Numeric ratio incorrect and explanation missing or incorrect (e.g. states absolute number of lines is a fixed property of the charge), or answer is blank/irrelevant.